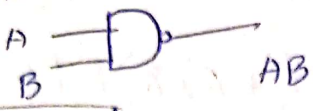
Assignment-2

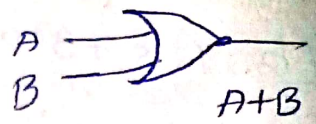
1. Realize all basic gate (AND, OR, NOT) using NAND and NOR separately?

Sol: NAND



A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

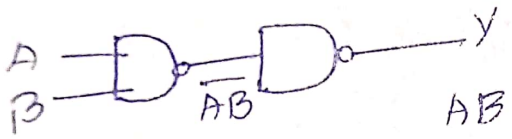
NOR



A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

(i) AND gate using NAND gate:-

→ complement of NAND gate is AND gate

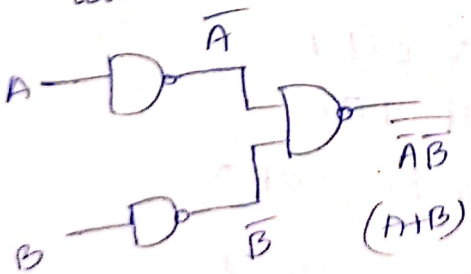


T.T

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

(ii) OR gate using NAND gate:-

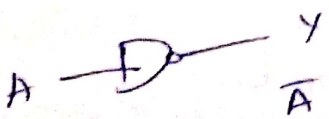
→ OR gate can be implement using De morgan law.



A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

(iii) NOT gate using NAND gate:-

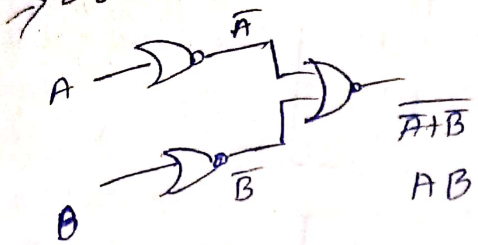
→ It can implemented by using NAND operation



A	Y
0	1
1	0

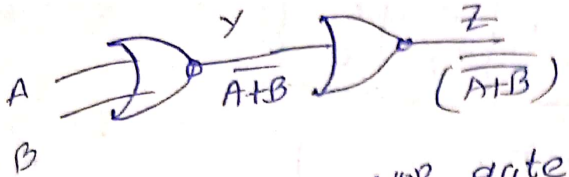
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→ By using De Morgan's Law



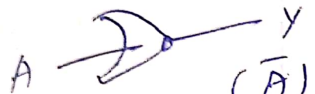
A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

v) OR gate using NOR gate:-
using De Morgan's Law



A	B	y
0	0	0
0	1	1
1	0	1
1	1	1

vi) NOT gate using NOR gate:-
It can be used with single i/p



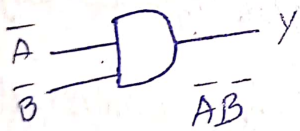
A	y
0	1
1	0

(2) Realize all the gates using Active low i/p:

Sol: Active low input means the signal is considered "ON" when logic '0' and OFF when logic is 1.

Truth tables:-

(i) AND:-



$\bar{A}$	$\bar{B}$	Y
0	0	1
0	1	0
1	0	0
1	1	0

$y = \overline{A}\,\overline{B}$

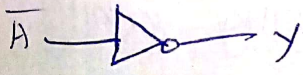
(ii) OR gate:-



$\bar{A}$	$\bar{B}$	Y
0	0	1
0	1	1
1	0	1
1	1	0

$y = \bar{A} + \bar{B}$

(iii) NOT gate:-

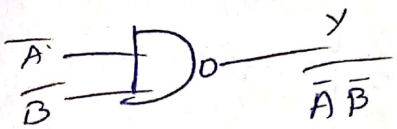


$\bar{A}$	Y
0	0
1	1

$y = \bar{A}$

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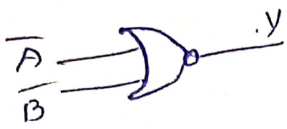
(iv) NAND gate:-



$y = (A+B)$

$\bar{A}$	$\bar{B}$	y
0	0	0
0	1	1
1	0	1
1	1	1

(v) NOR gate:-

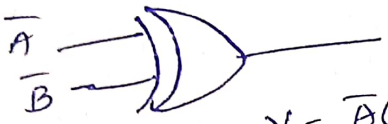


$\overline{A+B}$

$\bar{A}\bar{B}$

$\bar{A}$	$\bar{B}$	y
0	0	1
0	1	0
1	0	0
1	1	0

(vi) XOR gate:-

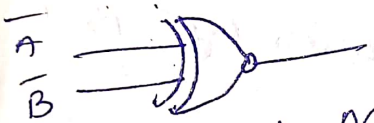


$y = \bar{A} \oplus \bar{B}$

$\bar{A} \odot B$

$\bar{A}$	$\bar{B}$	y
0	0	1
0	1	0
1	0	0
1	1	1

(vii) XNOR gate:-



$y = A \oplus B$

$\bar{A}$	$\bar{B}$	y
0	0	0
0	1	1
1	0	1
1	1	0

3. which gates are called universal gates? why

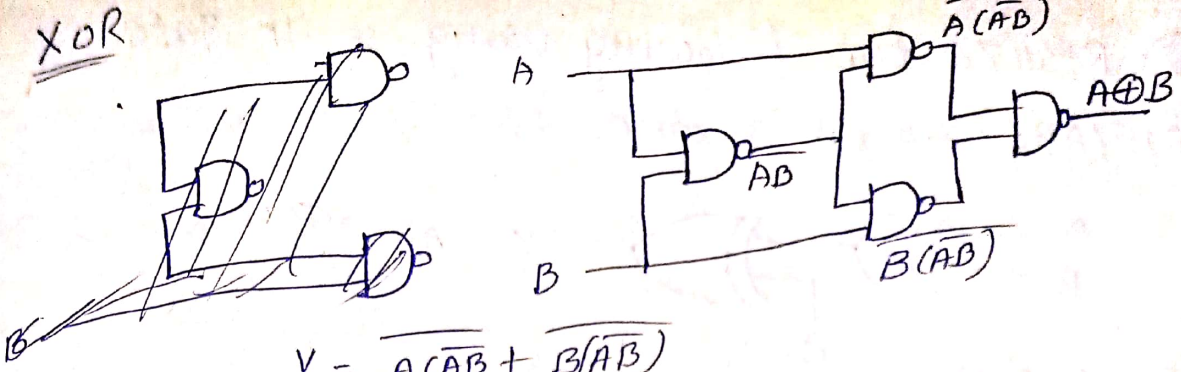
Sol:- NAND and OR gates are called as universal gates because that can be used to implement any boolean function without needing any other type of gates.

4. Give implementation of XOR using minimum number of NAND gates and show the implementation of XNOR gat using minimum number of NOR gates?

Sol:- To implement the both XOR & XNOR the minimum 4 NAND & NOR gates are required.

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XOR



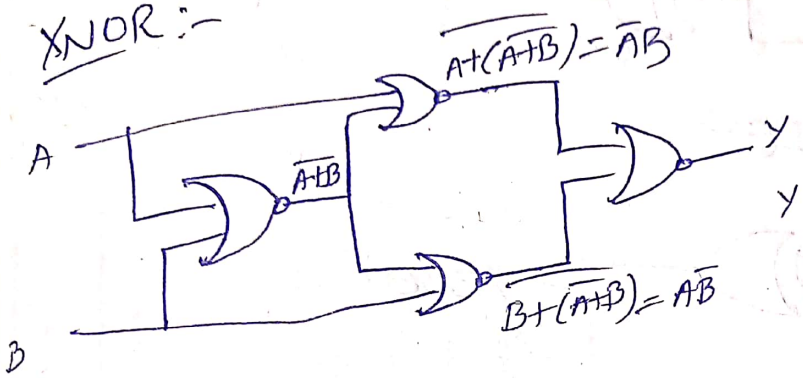
$Y = \overline{A(\overline{AB})} + \overline{B(\overline{AB})}$

$= \overline{A(\bar{A}+\bar{B})} + \overline{B(A+\bar{B})}$

$= \bar{A}+(AB)+\bar{B}+(AB)$

$(\bar{A}+A)B + (\bar{B}+B)A = \bar{A}B+\bar{B}A = A \oplus B$

XNOR :-



$\overline{A+(\bar{A}+B)} = (A+B)\cdot\bar{A}$

$A\bar{A}+\bar{A}B$

$= 0+\bar{A}B = \bar{A}B$

$y = \overline{\bar{A}B+A\bar{B}} = \overline{\bar{A}B}\cdot\overline{A\bar{B}}$

$= (A+\bar{B})(\bar{A}+B)$

$= A\bar{A}+AB+\bar{A}\bar{B}+B\bar{B}$

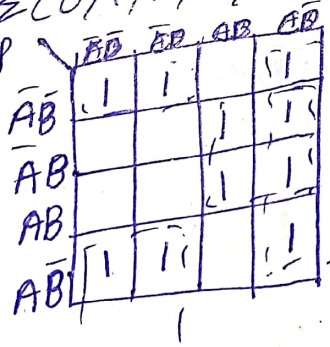
$= AB+\bar{A}\bar{B}$

$y = A \odot B$

(5) Simplify $F = \Sigma(0,6,1,2,7,8,9,10,14,15)$ in SOP and simplify the Boolean function $Y = \Sigma(0,2,3,5,7,10,11,15)$ in POS form using K-map.

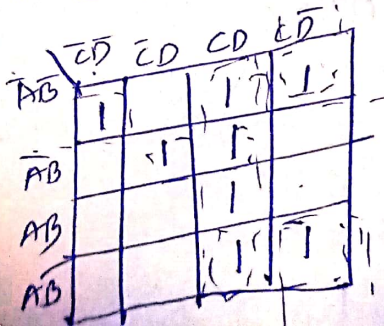
(i) $F = \Sigma(0,1,2,6,7,8,9,10,14,15)$

SOP



$F = \bar{B}\bar{C} + C\bar{D} + BC$

(ii) $F = \Sigma(0,2,3,5,7,10,11,15)$

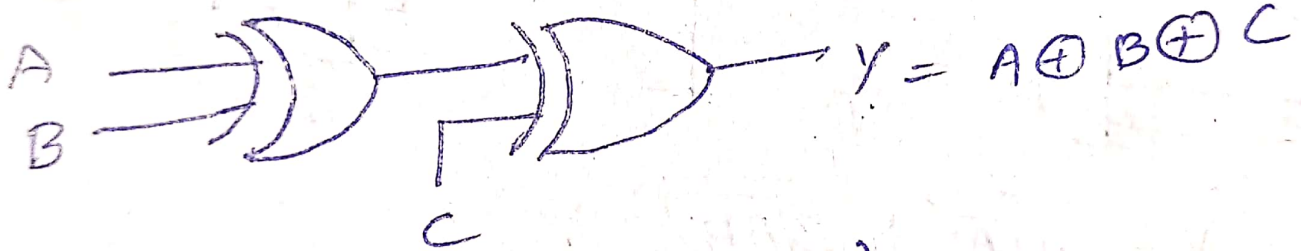


$F = (C+D)(\bar{D}+\bar{A}+\bar{B})(\bar{A}+B+D)\cdot(A+\bar{B}+C)$

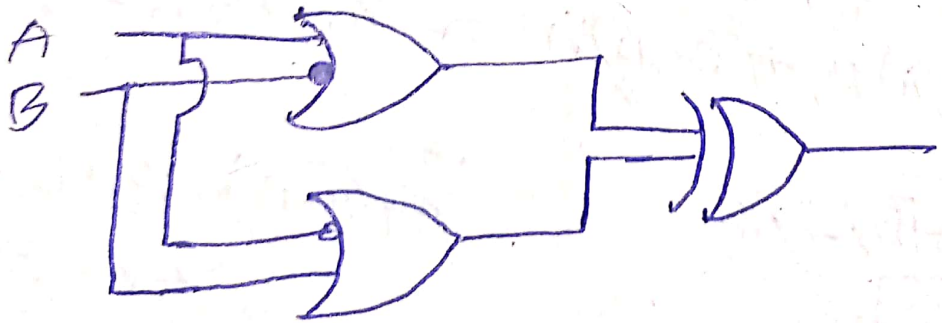
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(6) Realize the following using 2 i/p gate.

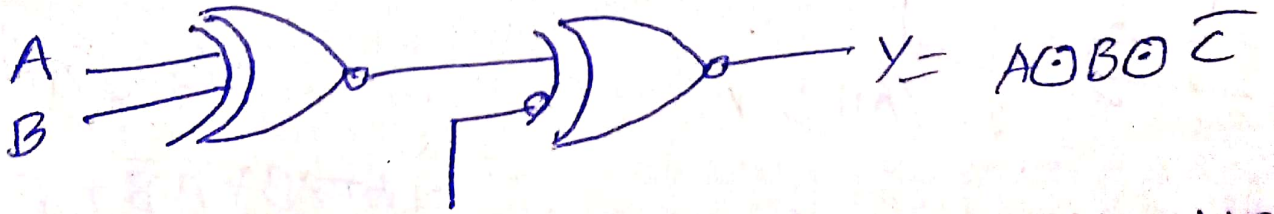
a) $F(ABC) = A \text{ XOR } B \text{ XOR } C$



b) $F(A,B) = (A + \bar{B}) \oplus (\bar{A} + B)$

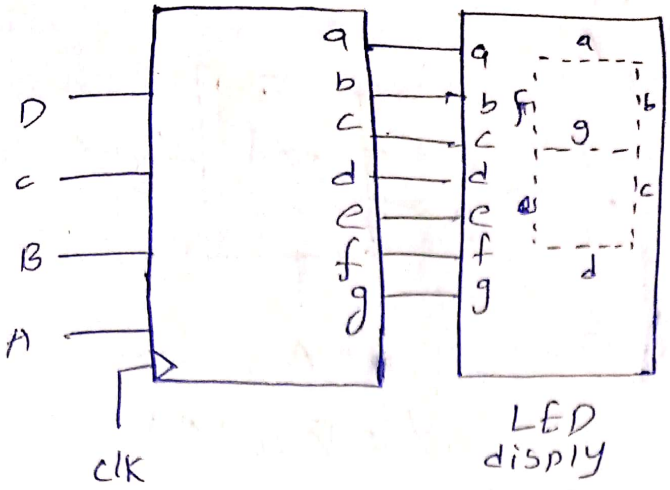


c) $F(ABC) = A \odot B \odot \bar{C}$



i/p NOR ga

(8) Design BCD to 7 segment display?

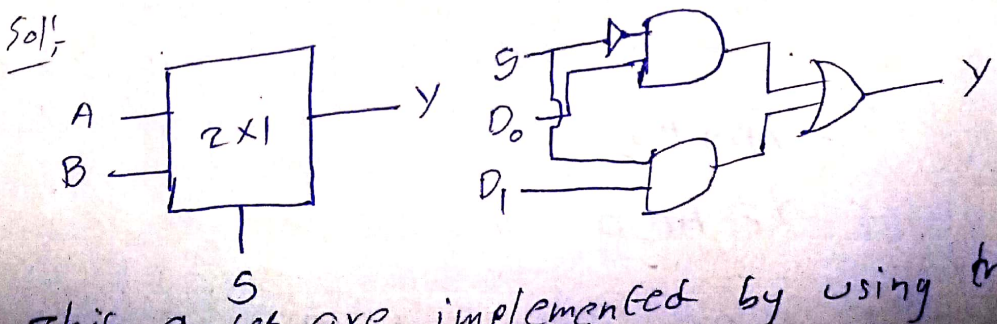


S	A	B	C	D	a	b	c	d	e	f	g
0	0	0	0	0	1	1	1	1	1	1	0
1	0	0	0	1	0	1	1	0	0	0	0
2	0	0	1	0	1	1	0	1	1	0	1
3	0	0	1	1	1	1	1	1	0	0	1
4	0	1	0	0	0	1	1	0	0	1	1
5	0	1	0	1	1	0	1	1	0	1	1
6	0	1	1	0	1	0	1	1	1	1	1
7	0	1	1	1	1	1	1	0	0	0	0
8	1	0	0	0	1	1	1	1	1	1	1
9	1	0	0	1	1	1	1	0	0	1	1

(9) Design the following gates using minimum number of 2:1 MUX

$y = \bar{S}D_0 + SD_1$

Sol:

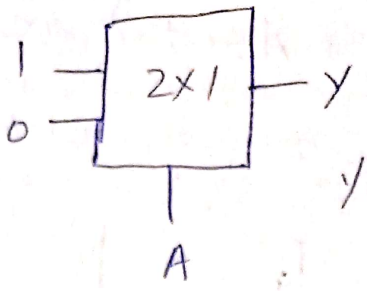


This gates are implemented by using truth table.

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a) NOT gate:-

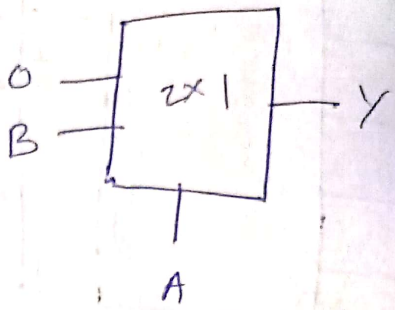
i/p	o/p
1	0
0	1



$Y = \bar{A}$

b) AND gate:-

i/p	o/p	Y	
0	0	0	0
0	1	0	
1	0	0	B
1	1	1	

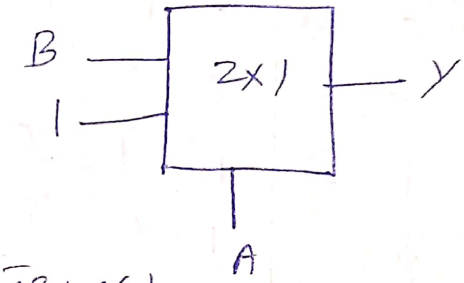


$Y = \bar{A}(0) + AB$

$Y = AB$

c) OR gate:-

i/p		o/p	Y	
0	0	0		
0	1	1	B	
1	0	1		1
1	1	1		



$Y = \bar{A}B + A(1)$

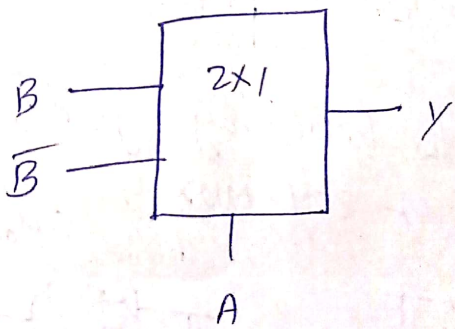
$Y = \bar{A}B + A$

$Y = (A+\bar{A})(A+B)$

$Y = A+B$

d) XOR gate:-

i/p		o/p	Y	
0	0	0	B	
0	1	1		
1	0	1	$\bar{B}$	
1	1	0		

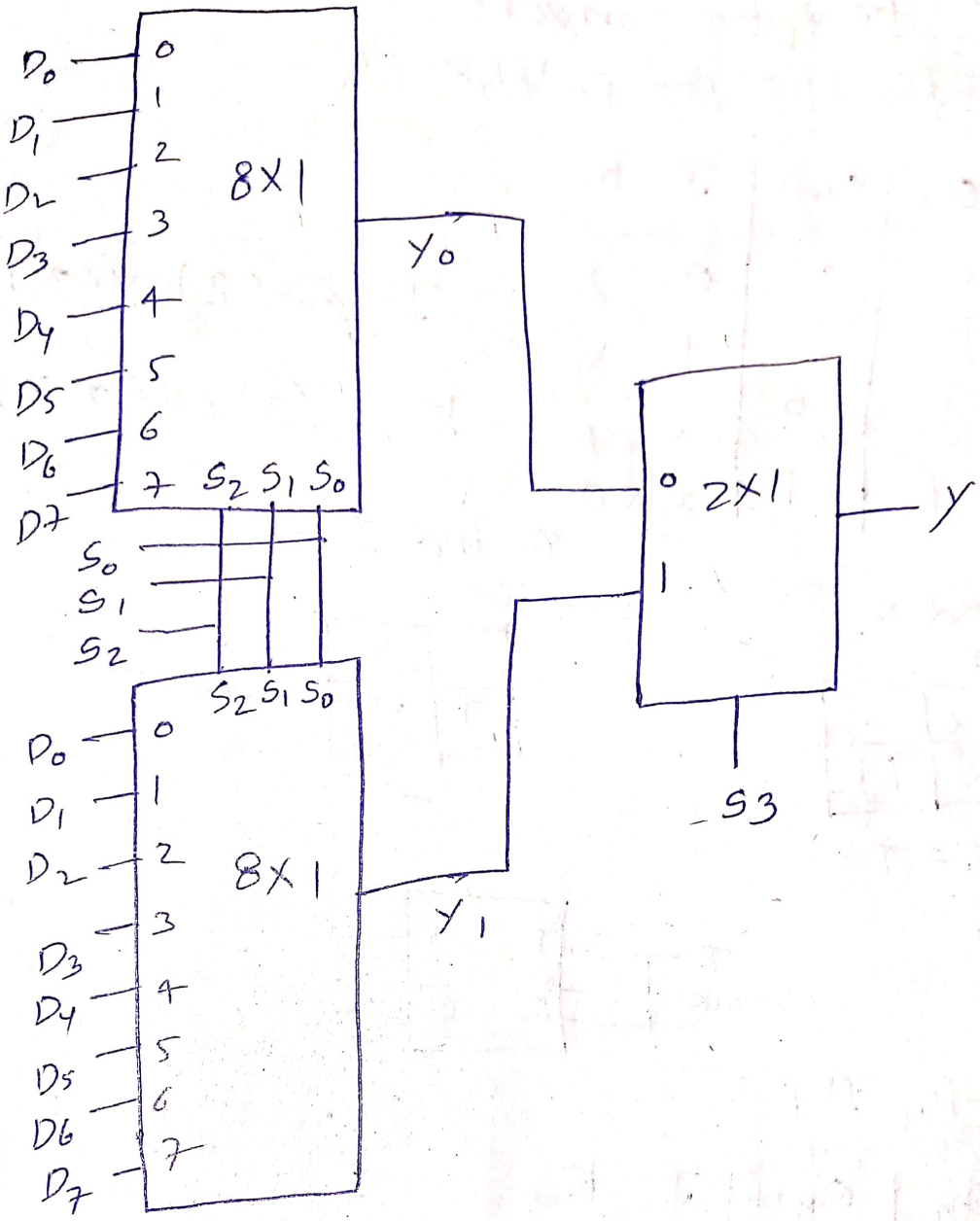


$Y = \bar{A}B + A\bar{B}$

$Y = A \oplus B$

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(15) Construct 16×1 MUX using two 8:1 MUX and one 2:1 MUX?



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